

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November – December 2011

EE 201 Control Engineering

Date: 01.12.2011, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Compare the Open loop and Closed loop system and give two example of each. [04]
- (b) Define the transfer function and also give the comparison of Block diagram and Signal Flow graph Method. [03]
- Q - 2 (a) Using Block diagram reduction technique for the system shown in fig.I, find the transfer function of the system. [06]

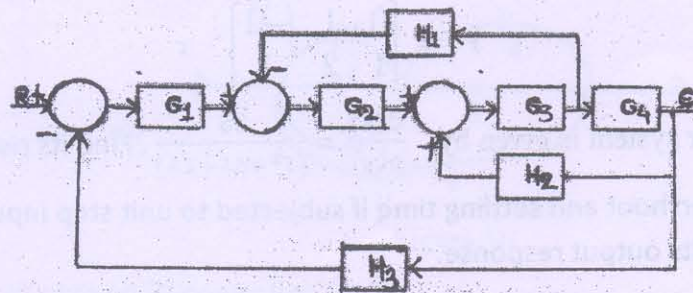


Figure.I

- (b) Obtain the transfer function C/R from the signal flow graph shown in figure. II [08]

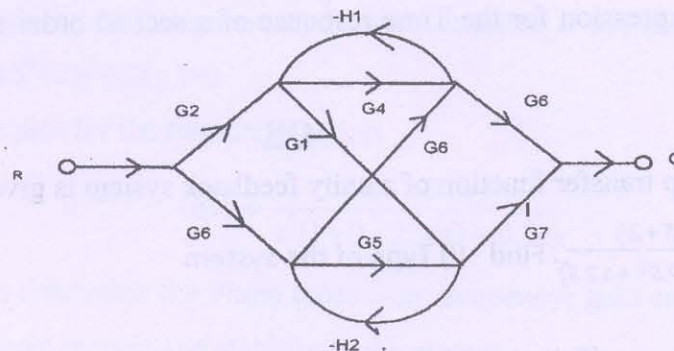


Figure.II

OR

Q-2 (a) Obtain the transfer function of the given state equation. [06]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = [0 \ 0 \ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(b) For the given system check the controllability and the Observability of the system. [08]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} * u$$

$$Y = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} \cdot x$$

Q-3 (a) A second order system is given by $\frac{C(S)}{R(S)} = \frac{25}{(S^2 + 6S + 25)}$. Find its rise time, peak [05]

time, Peak overshoot and settling time if subjected to unit step input. Also calculate expression for its output response.

(b) Measurement conducted on a servomechanism show the system response to be [04]

$$c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$$

When subjected to a unit step function.

Obtain the expression for the closed loop transfer function.

(c) Derive the expression for the Time response of a second order system with unit step input. [05]

OR

Q-3 (a) The open loop transfer function of a unity feedback system is given by [05]

$$G(S) = \frac{K(S+2)}{S(S^2 + 7S^2 + 12S)}. \text{ Find (i) Type of the system}$$

(ii) Static error coefficient and steady state error when input to the system is $\frac{R}{2} t^2$.

(b) What is steady state error? Explain static error coefficient method. [04]

(c) Consider the system as shown in fig.III [05]

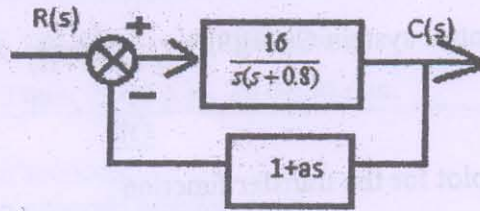


Figure.III

Determine the value of a that the damping ratio is 0.5. Also obtain the values of rise time and maximum overshoot M_p in its step response.

SECTION – II

Q - 4 (a) Determine the number of forward paths and the number of non-touching loops pairs for the signal flow graph in the figure VI are, respectively [04]

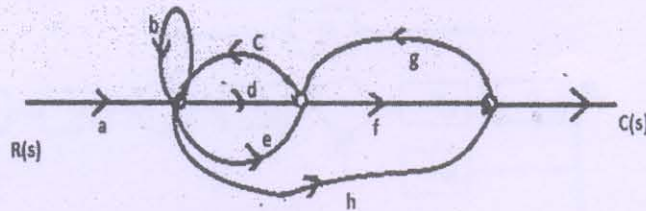


Figure.VI

(b) Write a short note on PI controller. [03]

Q - 5 (a) Examine the stability by Routh's criterion: $S^5 + S^4 + 2S^3 + 2S^2 + 3S + 15 = 0$. [06]

(b) Sketch the complete root locus of system having $G(S)H(S) = \frac{K}{s(s+1)(s+2)(s+3)}$. [08]

OR

Q - 5 (a) Find out the stability of the system by Routh's criterion if the characteristic equation is given by $S^4 + S^3 + 2S^2 + 2S + 3 = 0$ [06]

(b) Draw the Bode plot for the transfer function [08]

$$G(s) = \frac{16(1+0.5s)}{s^2(1+0.125s)(1+0.1s)}$$

From the graph determine the Phase cross over frequency; gain cross over frequency, phase margin, gain margin and stability of the system.

Q - 6 (a) Sketch the root locus for the system having $G(S)H(S) = \frac{K(S+4)}{S(S^2+2S+2)}$. [07]

(b) For a certain control system $G(S)H(S) = \frac{K}{S(S+2)(S+10)}$. Sketch the Nyquist plot. [07]

OR

Q - 6 (a) Draw the Bode plot for the transfer function [07]

$$G(s) = \frac{1000}{s(1+0.1s)(1+0.001s)}$$

From the graph determine the Phase cross over frequency; gain cross over frequency, phase margin, gain margin and stability of the system.

(b) Sketch the nyquist plot for a system with $G(S)H(S) = \frac{10(S+1)}{(1+2S)(1+0.1S)(1+0.02S)}$. [07]

Comment on its stability.
